

General Objective

The project study aimed to develop **LeaveSync: A Leave Management System with Device Fingerprinting and Digital Signature-Based Approval for The Lewis College** to provide a secure, automated, and non-repudiable platform for managing employee leave credits and approvals.

Specific Objectives

The specific objectives of the study were:

1. **To develop a web-based leave management module** that centralizes the submission, tracking, and filing of employee leave requests to replace manual, paper-based workflows.
2. **To integrate a device fingerprinting security layer** that captures and validates specific hardware and network attributes including browser type, OS version, and IP patterns to ensure that only registered and trusted devices can access the system.
3. **To implement a digital signature-based approval process** using cryptographic signing to verify manager identities, ensuring the authenticity of approvals and providing non-repudiation for all leave transactions.
4. **To conduct automated black-box and security testing** using tools like **Selenium** and **OWASP ZAP** to evaluate the system's functional accuracy and the effectiveness of the device-aware verification measures.